The first law

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use a consistent sign convention in an energy balance.

SUCCESS CRITERIA - I can...

1. State whether heat enters or leaves.
2. Distinguish work done by the system from work done on it.
3. Apply ΔU=Q-W with the stated signs.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A 3 kg sample completely changes phase with latent heat 500 J/kg. Find the energy magnitude required.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $\Delta U = Q - W_{by}$.
Q is positive into the system; W_{by} is positive when the system does work on its surroundings.

Worked reasoning: A system receives 90 J of heat and does 30 J of work on its surroundings. With ΔU=Q-W, internal energy increases by 60 J.

Your turn: Draw labeled arrows crossing a system boundary and mark the 60 J stored increase.

Draw and label all energy transfers across the boundary.

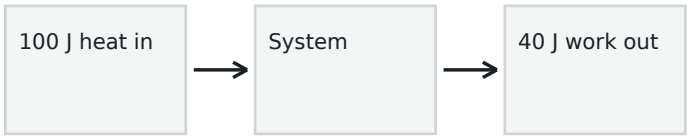
Chosen system

Stored energy change: _____ J

Marzano check - circle: ① ② ③

03 Read and interpret the model

A system absorbs 100 J of heat and does 40 J of work. Draw the energy arrows and find its internal-energy change.



Marzano check - circle: ① ② ③

04 Apply the physics

A gas in a piston absorbs 80 J of heat and does 20 J of work by pushing outward. Label the input, work output and stored-energy change. Use $\Delta U = Q - W_{by}$ to find whether its internal energy increases or decreases.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: ① ② ③

The first law

antocciphysics

Continue down each column.

05 Explain your reasoning

Why does the phrase “work is 60 J” need a direction or sign convention?

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Total kinetic energy changes from 24 J to 20 J in an isolated collision. Is it elastic? Where could the difference go?
Show the energy changes and transfers; label units.

Initial **Transfers** **Final**



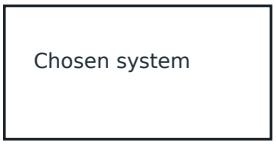
Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

07 Exit check

Heat input 30 J; work output 10 J. Find ΔU. Label energy entering, leaving or stored on the system model.

Draw and label all energy transfers across the boundary.



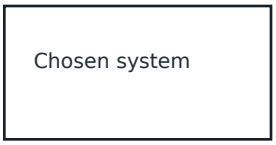
Stored energy change: _____ J

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Give two different heat-input/work-output pairs that increase internal energy by 40 J. Label energy entering, leaving or stored on the system model.

Draw and label all energy transfers across the boundary.



Stored energy change: _____ J

Marzano check - circle: 1 2 3